RESEARCH ATLAS • 05 NOVEMBER 2026

History belongs on the chart, not in an unstructured list

Historical market events | A source-linked tree for rates, bonds, FX and portfolio context

CENTRAL CONCLUSION

Build one reusable event library, with a hierarchical tree for browsing and independent tags for filtering. Keep each dated milestone connected to its source and to an honest data frequency.

The architecture is simple; the evidence discipline is the work.



Design recommendation. Counts describe this report's selected starter catalogue, not the universe of historical crises.

01 Group by mechanism, not by headline alone. A primary folder helps navigation; it is not proof that an event had only one cause.

02 Keep episodes and milestones separate. A multi-year crisis is not a one-day announcement.

03 Use more than one time scale. Monthly averages explain context; daily observations expose abrupt moves.

04 Show facts, interpretation and gaps separately. A marker beside a price move does not establish causation.

05 Implement in stages. Start with verified records and source links before adding rankings, filters or portfolio simulations.

Scope: selected episodes from 1929 to April 2025. This is a historical research and design report, not a current market bulletin, trading recommendation or deployed chart feature. No client information is used.

READER MAP

Read the report in three passes

Start with the tree, inspect the evidence, then use the implementation blueprint.

A report that can become a maintainable feature



Design map, not a delivery timetable.

Section	Pages
Architecture and style	3–8
Verified historical catalogue	9–12
Comparisons and worked cases	13–22
Chart and implementation blueprint	23–27
Official source register	28–31

The report contains real historical observations, not invented market paths. Four monthly windows, one daily FX zoom and one policy-rate step chart illustrate the annotation method. Other diagrams describe mechanisms and are explicitly non-numerical.

The companion files make the work reusable. The HTML is an offline, responsive reading version; the JSON is a research starter; the CSV retains source precision. They are not integrated into an existing website or connected to a live feed.

> **EVIDENCE KEY**
>
> REPORTED = source-backed historical statement. ANALYSIS = interpretation or design recommendation. MISSING = unavailable observation, never zero. A source link is evidence for the stated claim, not a blanket endorsement of everything written nearby.

01 • INFORMATION ARCHITECTURE

Use five levels, but store each fact only once

The tree is the route into the library, not the database itself.

From broad grouping to a defensible chart marker



Illustrative hierarchy using a sourced milestone [S30]. Family and mechanism placement are editorial recommendations.

A level-five annotation should reference a milestone ID, not copy its facts. The same policy decision can appear on a gilt-yield chart and an FX chart, while its date and source remain maintained in one place. Correct the underlying record once and every linked view can update.

One episode can contain different kinds of milestone. Trigger, market dislocation, institution failure, policy announcement, policy effective date and programme end are separate types. Do not force a “recovery” marker unless recovery is defined for a particular series and horizon.

Sources: S30. Design interpretation is this report’s recommendation.

02 • EVENT TREE, PART A

Group financial stress by its dominant mechanism

The six families are proposed navigation categories, not new colour assignments.

Rates & monetary regimes

Level 2: mechanism	Level 3: historical episode
Currency-regime changes	↳ End of dollar–gold convertibility (1971) ↳ Sterling leaves the ERM (1992) ↳ Swiss franc floor removal (2015)
Monetary tightening and inflation	↳ Volcker operating-framework change (1979–1982) ↳ Fed tightening cycle (2022–2023)
Policy communication and term repricing	↳ Taper tantrum (2013)

Credit, banks & sovereign stress

Level 2: mechanism	Level 3: historical episode
Bank runs and deposit confidence	↳ US banking panics (1930–1933)
Asset–liability mismatch	↳ Savings and loan crisis (1980s–early 1990s)
Foreign-currency debt and rollover stress	↳ Asian financial crisis (1997–1998)
Housing losses and funding contagion	↳ Global financial crisis (2007–2009)
Sovereign–bank feedback	↳ Euro-area sovereign stress (2010–2012)
Duration exposure and concentrated deposits	↳ Silicon Valley Bank failure (2023)
Confidence and bank funding	↳ Credit Suisse rescue (2023)

WHY THE UK GILT EPISODE SITS HERE

The fiscal announcement is a trigger; margin calls and forced deleveraging describe the amplification. Keep both in tags, but keep only one canonical episode record.

Proposed classification. Detailed evidence and dated anchors follow in the catalogue. Family labels remain neutral; approved entity and asset colours retain their existing meanings.

03 • EVENT TREE, PART B

Separate forced liquidation from valuation repricing

These branches focus on transmission: how a market disturbance becomes harder to absorb.

Funding, leverage & liquidity

Level 2: mechanism	Level 3: historical episode
Leveraged convergence trades	↳ Russia shock and LTCM (1998)
Short-term funding scarcity	↳ US repo-market stress (2019)
Margin calls and forced deleveraging	↳ UK gilt / LDI crisis (September–October 2022)
Crowded funding trades and deleveraging	↳ August carry-trade unwind (August 2024)

Valuation & market structure

Level 2: mechanism	Level 3: historical episode
Equity repricing and trading stress	↳ Wall Street crash (1929)
Mechanical selling and market structure	↳ Black Monday (1987)
Valuation and earnings expectations	↳ Dot-com reversal (2000–2002)

A price fall does not identify its own mechanism. A valuation reset can occur without a funding run. A financing constraint can force a sale even when an investor continues to believe the asset is worth more. The catalogue should make those questions visible rather than assign every episode a generic crash label.

> **KEEP SECONDARY PATHS SEARCHABLE**
>
> An episode placed under liquidity can still carry equity, FX, policy and regional tags. The primary path is an editorial route into the evidence, not an exclusion of other channels.

03 • EVENT TREE, PART C

Keep external shocks and political policy visible

A long list becomes useful when the branches answer different questions.

Real-economy & external shocks

Level 2: mechanism	Level 3: historical episode
Energy supply shocks	↳ Oil embargo and supply shock (1973–1974)
Domestic economic adjustment	↳ Singapore recession (1985)
Operational and payment disruption	↳ September 11 disruption (2001)
Public-health disruption	↳ SARS (2003)
Public-health shock and demand for cash	↳ COVID-19 market crisis (2020)

Fiscal, trade & political policy

Level 2: mechanism	Level 3: historical episode
Trade-policy announcements	↳ April tariff shock (April 2025)

Link neighbouring episodes without merging them. The 1929 crash and the early-1930s banking panics are related but distinct records. Likewise, COVID-19 is a broad episode with a market-liquidity phase; it does not need a duplicate copy in every branch.

Coverage should expand without breaking the IDs. Japan’s asset-price collapse, the 1994 bond sell-off, the 2010 flash crash and Brexit are sensible future additions, but are not presented as source-audited entries in this starter catalogue.

> **PRACTICAL RULE**
>
> Every episode has one primary path. Secondary mechanisms and related episodes are links or tags. A search result must deduplicate by the canonical episode ID, not by the visible title.

04 • COLOUR, TAGGING & TYPOGRAPHY

Use the tree for browsing and tags for comparison

One entity keeps one approved colour, even when it appears in several analytical views.

Independent axis	What it answers	Example
Region	Where is the exposure or event?	Singapore / US / Switzerland
Person	Who made the statement?	Only a verified named person
Institution	Who acted?	Federal Reserve / SNB / MAS
Asset	What instrument or currency?	US Treasuries / USD/CHF
Sector	What activity is exposed?	Banks / energy
Company or ticker	Which issuer or security?	Resolve identity before tagging
Macro indicator	What is measured?	Treasury yields / credit spreads
Event	What happened?	Policy decision / liquidity stress
Editorial status	What is the evidence state?	REPORTED / ANALYSIS / UNVERIFIED

Follow the workbook's actual matching order. Validate context first, choose the longest valid whole phrase, then use priority for a genuine equal-span conflict. Do not match "US" inside "USD" or "Fed" inside "FedEx". Ambiguous aliases remain unresolved until the context establishes identity.

Keep direction separate from identity. A bond term can remain Bonds Green while its price falls. A yield statistic uses Rates Blue rather than automatically inheriting the bond-instrument colour. Evidence uncertainty uses Status Gray, not a loss signal.

> **FONT AND LAYOUT APPLICATION**
>
> This report uses DejaVu Sans and DejaVu Sans Bold, matching the fonts embedded in the supplied visual guide. The guide requires readable hierarchy rather than prescribing a named family. The HTML uses a system fallback and 16px body text; PDF tables retain labels and units.

Style authority: Market_Colour_Tag_Master_v1_2_Merged.xlsx; Colour_Tag_Visual_Design_Guide_v1_2_Merged.pdf, especially pages 6–8, 17 and 22–23. No new keyword-to-colour mappings are introduced.

05 • DATE AND EVIDENCE DISCIPLINE

A historical event has more than one relevant date

A precise-looking timestamp is not useful when it describes the wrong thing.

Date field	Meaning	Example or rule
Event / announcement	When the development was announced or occurred	Fed decision: 16 March 2022
Effective date	When the operational change took effect	Target range: 17 March 2022
Market observation	When the displayed price or rate was observed	H.10: New York noon, not session close
Publication date	When the source was released	SVB review: 28 April 2023
Retrieval date	When the source was checked for this report	5 November 2026
Date precision	Day, month, year or broader period	S&L; crisis: multi-year episode
Display / study window	Chosen analytical period around an anchor	12 months before and after, where available

Preserve local time and the trading calendar. Store an exact time and time zone only when supported. A Sunday announcement can influence Monday trading; after-close announcements can belong economically to the next session. Date-only records should remain date-only, not silently become midnight UTC.

A monthly label is not an executable price at the start of the month. FRED’s monthly Treasury observations are averages. The month label should be displayed as a period, and an event-month band should not pretend to identify the announcement-day reaction.

Do not shift an event to make the chart look more convincing



Reported policy dates [S28–S29]; observation rule is a design recommendation.

Sources: S23, S28, S29, S31, S36.

06 • VERIFIED STARTER CATALOGUE 1/4

1929–1985: crashes, regimes and structural stress

Selected anchors are not claimed to be the first cause or final end of each episode.

1929-10-28 | Wall Street crash

Fact. The late-October stock-market crash preceded the banking crises of the early 1930s. [S01]

Chart use / analysis: Link related episodes, but do not collapse the crash and every later bank failure into one daily marker. **Precision:** day.

1933-03-06 | US banking panics

Fact. The national banking holiday began on 6 March 1933 amid the banking crisis. [S02]

Chart use / analysis: Use a broad episode plus specific policy milestones; the holiday is not the start of the whole Depression. **Precision:** day.

1971-08-15 | End of dollar–gold convertibility

Fact. The US suspended dollar convertibility into gold on 15 August 1971. [S03]

Chart use / analysis: Separate the announcement from the wider transition away from Bretton Woods. **Precision:** day.

1973-10 | Oil embargo and supply shock

Fact. The October 1973 oil shock is a supply-side episode, not a single universal market-close event. [S04]

Chart use / analysis: Pair commodity, inflation and activity charts; do not assume all asset prices respond in the same direction. **Precision:** month.

1979-10-06 | Volcker operating-framework change

Fact. On 6 October 1979 the Fed changed operating procedures in its anti-inflation effort. [S05]

Chart use / analysis: Mark the regime change separately from later interest-rate peaks or recession dates. **Precision:** day.

1980s–early 1990s | Savings and loan crisis

Fact. Savings institutions faced serious funding and asset-side vulnerabilities over a multi-year crisis. [S06]

Chart use / analysis: Do not invent a 1 January trigger. Use sourced resolution milestones when expanding the catalogue. **Precision:** period.

1985 | Singapore recession

Fact. Singapore’s 1985 recession reflected external weakness and domestic structural pressures. [S07]

Chart use / analysis: A year-level label is more honest than a fabricated daily market trigger. **Precision:** year.

06 • VERIFIED STARTER CATALOGUE 2/4

1987–2003: trading, currencies and external disruption

Selected anchors are not claimed to be the first cause or final end of each episode.

1987-10-19 | Black Monday

Fact. Black Monday occurred on 19 October 1987; the Fed issued a liquidity-support statement the next day. [S08]

Chart use / analysis: Display the crash and response separately; a crash is not automatically a banking collapse. **Precision:** day.

1992-09-16 | Sterling leaves the ERM

Fact. The UK suspended sterling's ERM membership on 16 September 1992. [S09]

Chart use / analysis: A defended exchange-rate arrangement should not be treated as an unconditional floor. **Precision:** day.

1997-07-02 | Asian financial crisis

Fact. Thailand changed the baht exchange-rate regime on 2 July 1997 as regional financial stress developed. [S10]

Chart use / analysis: Study currency denomination, maturity mismatch and refinancing together. **Precision:** day.

1998-09-23 | Russia shock and LTCM

Fact. A private-sector recapitalisation of LTCM was arranged on 23 September 1998 with New York Fed facilitation. [S11]

Chart use / analysis: Forced liquidation can overwhelm apparently diversified positions; facilitation is not a Fed-funded bailout. **Precision:** day.

2000-03 | Dot-com reversal

Fact. Major US equity indexes reached peaks in March 2000 before the subsequent reversal discussed by the Fed. [S12]

Chart use / analysis: Use source-appropriate month precision unless a chosen index's exact peak is separately verified. **Precision:** month.

2001-09-11 | September 11 disruption

Fact. The attacks disrupted financial operations and payment flows, prompting emergency liquidity support. [S13]

Chart use / analysis: Distinguish exchange closures and operational liquidity from normal traded price observations. **Precision:** day.

2003-03-12 | SARS

Fact. WHO issued a global alert on 12 March 2003. [S14] [S15]

Chart use / analysis: This alert is an annotation milestone, not the first infection or a universal start date for economic damage. **Precision:** day.

06 • VERIFIED STARTER CATALOGUE 3/4

2007–2020: credit, communication and liquidity

Selected anchors are not claimed to be the first cause or final end of each episode.

2008-09-15 | Global financial crisis

Fact. Lehman Brothers filed for bankruptcy on 15 September 2008 during the wider financial crisis. [S16] [S17]

Chart use / analysis: Separate institutional failures, emergency support and subsequent policy easing. **Precision:** day.

2012-09-06 | Euro-area sovereign stress

Fact. The ECB announced the technical features of conditional Outright Monetary Transactions on 6 September 2012. [S19]

Chart use / analysis: An announced conditional backstop is not the same thing as a completed purchase programme. **Precision:** day.

2013-05-22 | Taper tantrum

Fact. May 2013 testimony and subsequent June communication were interpreted as signalling future asset-purchase tapering. [S20]

Chart use / analysis: Market expectations can move before the operational policy change. **Precision:** day.

2015-01-15 | Swiss franc floor removal

Fact. The SNB discontinued its CHF 1.20 per euro minimum exchange rate on 15 January 2015. [S21] [S22]

Chart use / analysis: The policy instrument was EUR/CHF; a USD/CHF chart measures a different cross. **Precision:** day.

2019-09-17 | US repo-market stress

Fact. Repo-market stress intensified on 17 September 2019 and the Fed intervened. [S25]

Chart use / analysis: Money-market stress is not automatically a sovereign credit event. **Precision:** day.

2020-03-15 | COVID-19 market crisis

Fact. The Fed announced a 0–0.25% target range on Sunday, 15 March 2020. [S26]

Chart use / analysis: Separate the health shock, market dysfunction and policy actions inside the same episode. **Precision:** day.

06 • VERIFIED STARTER CATALOGUE 4/4

2022–2025: inflation, leverage and policy shocks

Selected anchors are not claimed to be the first cause or final end of each episode.

2022-03-16 | Fed tightening cycle

Fact. The Fed announced the first increase of this tightening sequence on 16 March 2022. [S28] [S29]

Chart use / analysis: Keep announcement dates, effective dates and the plotted rate boundary explicit. **Precision:** day.

2022-09-23 | UK gilt / LDI crisis

Fact. The UK fiscal statement on 23 September preceded severe gilt-market dysfunction. [S30]

Chart use / analysis: The primary folder describes amplification through leverage, not an exclusive explanation of the trigger. **Precision:** day.

2023-03-10 | Silicon Valley Bank failure

Fact. SVB failed on 10 March 2023; the Fed's review describes management and supervisory weaknesses. [S31]

Chart use / analysis: Analyse interest-rate exposure, liquidity and depositor concentration together. **Precision:** day.

2023-03-19 | Credit Suisse rescue

Fact. The UBS takeover and SNB liquidity support were announced on 19 March 2023. [S32]

Chart use / analysis: Keep the Sunday announcement distinct from the next trading session and later acquisition completion. **Precision:** day.

2024-08-05 | August carry-trade unwind

Fact. BIS identifies early-August turbulence and the 5 August peak, amplified by carry-trade unwinding. [S33]

Chart use / analysis: Do not attribute every move to one trade or treat uncertain total positioning as measured fact. **Precision:** day.

2025-04-02 | April tariff shock

Fact. The 2 April tariff announcements formed a major part of the IMF's April 2025 risk assessment. [S34] [S35]

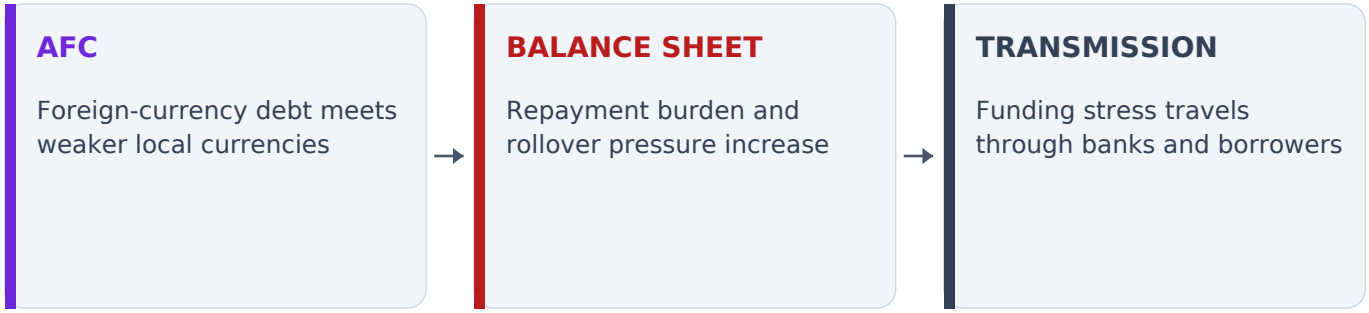
Chart use / analysis: Policy-announcement, implementation and revision dates belong in separate milestones. **Precision:** day.

07 • COMPARATIVE CASE: AFC AND LTCM

Currency mismatch and forced selling are different risks

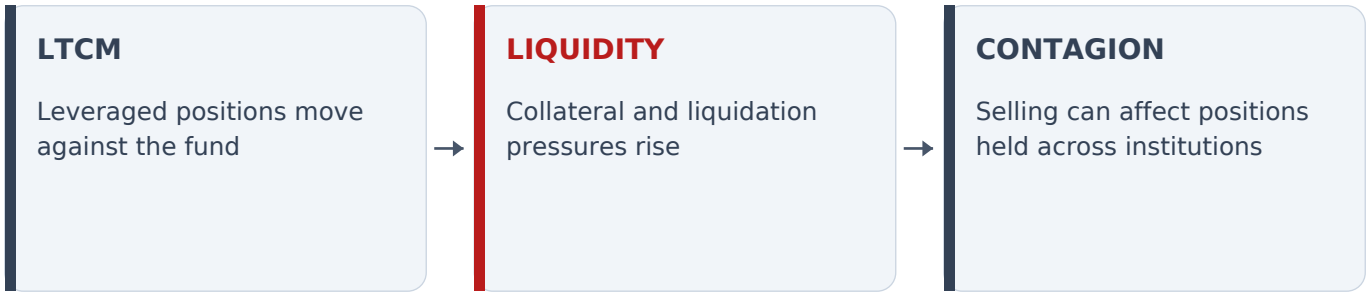
Two episodes can damage many markets without sharing the same balance-sheet weakness.

AFC: start with the currency and maturity of the liability



Conceptual transmission diagram, not a numerical path. Historical context [S10]; causal decomposition is analysis.

LTCM: start with leverage, position crowding and exit capacity



Conceptual transmission diagram. Historical chronology and private recapitalisation [S11].

The borrowing rate is only one part of a financed position. For analytical comparison, inspect currency mismatch, refinancing dates, collateral rules and how quickly assets can be sold. These are distinct from an asset's expected long-term return; a favourable return thesis does not eliminate a near-term funding constraint.

A policy response does not guarantee protection for every investor. The LTCM recapitalisation was private-sector funded with New York Fed facilitation. Do not code it as a direct Fed capital injection, and do not turn an historical intervention into an assumed rescue rule for future portfolios.

CHART ACTION

Use linked but separate panels: asset price or spread, funding conditions, FX and policy milestones. Avoid compressing all four into an unlabeled "crisis severity" number.

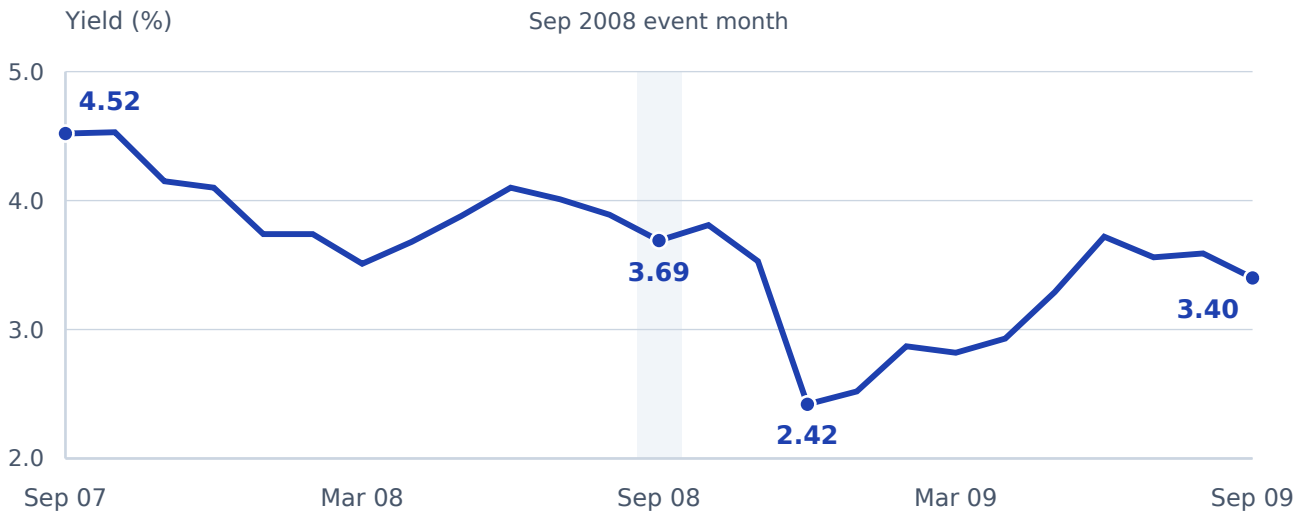
Sources: S10, S11.

08 • WORKED CHART: GLOBAL FINANCIAL CRISIS

GFC yields fell overall, but not in a straight line

A monthly view gives context around September 2008; it does not isolate Lehman’s daily effect.

US 10-year Treasury yield | September 2007–September 2009



Source: FRED GS10 / Federal Reserve Board [S36]. Unit: percent; monthly business-day average. Shaded band marks September 2008. How to read: follow the line left to right; the vertical scale begins at 2%, not zero. Retrieved 5 November 2026.

Observation	Yield	What the comparison does and does not show
September 2007	4.52%	Context 12 months before the event month
September 2008	3.69%	Average contains observations before and after Lehman
October 2008	3.81%	Higher than September despite crisis stress
December 2008	2.42%	Later lower yield; not a single-day event return
September 2009	3.40%	12-month-after context, not a crisis-end definition

The intermediate path matters. September to October 2008 shows a rise of 12 basis points, even though the broader period includes a large decline. A two-point “before versus after” comparison can conceal the sequence that matters to an investor facing collateral calls or an urgent sale.

The milestone layer must remain distinct from the data. September 15 marks Lehman’s filing; September 16 marks AIG support; December 16 marks the Fed’s 0–0.25% target-range announcement. These do not convert a monthly yield chart into a causal estimate.

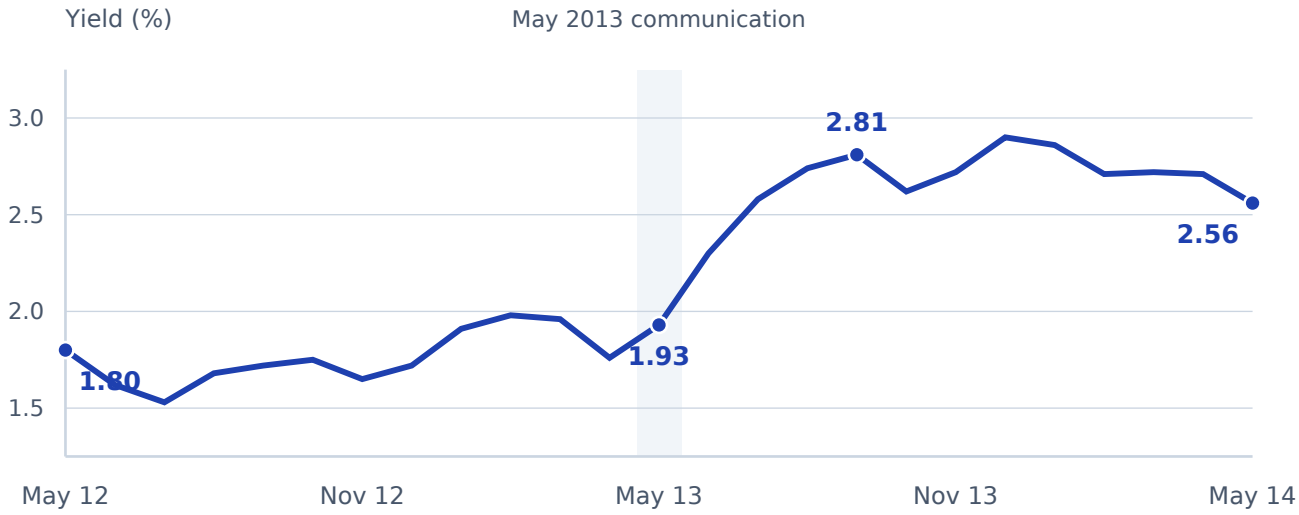
Sources: S16, S17, S18, S36. Yield differences calculated from the displayed source observations; a yield change is not a bond total return.

09 • WORKED CHART: TAPER TANTRUM

Markets can reprice a policy before it happens

Communication, implementation and market anticipation belong in different fields.

US 10-year Treasury yield | May 2012–May 2014



Source: FRED GS10 [S36]. Unit: percent; monthly business-day average. Event-month band: May 2013. How to read: compare the rise in subsequent monthly averages, not a fabricated one-day reaction. Non-zero scale; retrieved 5 November 2026.

OBSERVED CHANGE

May 2013: 1.93%. September 2013: 2.81%. Difference = (2.81 − 1.93) × 100 = +88 basis points. This is a monthly-average yield change, not an 88% move and not a bond return.

The useful event is a sequence of communications. The Fed's retrospective research identifies May testimony and June communication as important to the market's interpretation of future tapering. The chart should not label the May milestone "QE ended", because that would confuse expectations with implementation.

The same announcement can transmit through several assets. A Treasury chart shows one channel; emerging-market currencies and funding conditions show others. The classification belongs under policy communication, with FX and region tags available as secondary filters rather than duplicate event entries.

Analytical chain: communication → revised expectations → market repricing. Neither the direction nor the size of the reaction is assumed for every announcement.

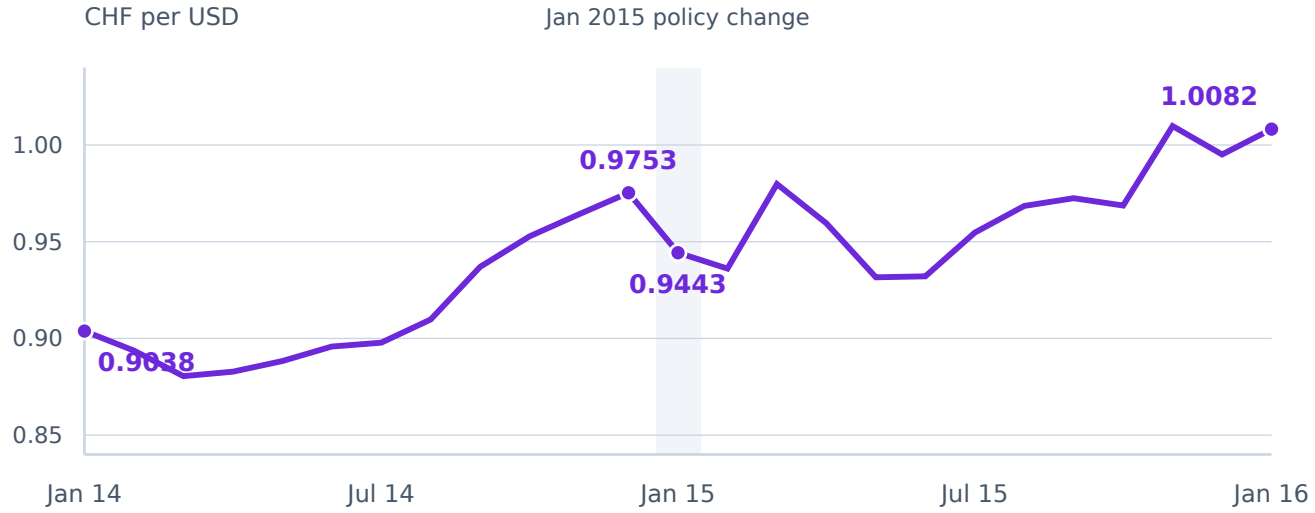
Sources: S20, S36.

10 • WORKED CHART: SWISS FRANC, CONTEXT VIEW

A monthly CHF chart can hide a severe daily shock

Keep the long window, but do not let averaging erase the event.

CHF per USD | January 2014–January 2016



Source: FRED EXSZUS / Federal Reserve Board [S37]. Monthly averages; units CHF per USD. How to read: a lower line means fewer Swiss francs buy one US dollar, so the Swiss franc is stronger against USD. Non-zero scale. Retrieved 5 November 2026.

Month	CHF per USD	Interpretation
December 2014	0.9753	Monthly average before the announcement month
January 2015	0.9443	Mixes observations from both sides of the policy change
February 2015	0.9361	Another monthly average, not the announcement-day low
January 2016	1.0082	Later context cannot undo an earlier forced sale

The policy anchor and the plotted pair are not identical. The SNB removed a minimum exchange rate of CHF 1.20 per euro on 15 January 2015. The figure above is USD/CHF, chosen for funding analysis; it should never be titled as the direct EUR/CHF floor itself.

A smooth-looking line is not evidence of a smooth experience. A monthly mean combines rates before and after an abrupt change. The next page therefore changes frequency, not the historical facts. A production chart should make that frequency switch obvious and preserve both views.

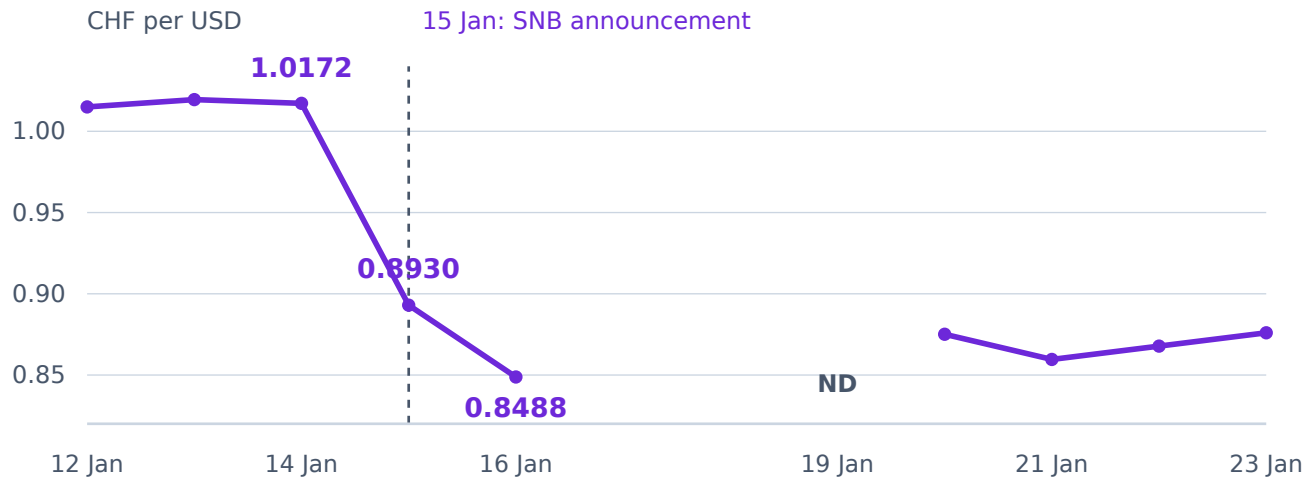
Sources: S21, S22, S37.

11 • WORKED CHART: SWISS FRANC, DAILY VIEW

FX quote direction changes the liability calculation

Use the quote’s units before interpreting the sign of a return.

Daily CHF per USD | 12–23 January 2015



Sources: Federal Reserve H.10 releases [S23–S24]. New York noon buying rates, not intraday highs/lows or executable quotes. How to read: lower CHF per USD means a stronger CHF. ND on 19 January remains missing, so the line is broken. Retrieved 5 November 2026.

Exact calculation, 14 to 15 January	Result
CHF per USD: (0.8930 / 1.0172 − 1) × 100	−12.209988...%
USD value of a fixed CHF amount: (1.0172 / 0.8930 − 1) × 100	+13.908174...%
SGD per CHF uses (SGD per USD) / (CHF per USD)	Cross-rate, not subtraction

The reciprocal is the important distinction. Let L be an unchanged CHF liability and q be CHF per USD. Its USD equivalent is L ÷ q. Comparing the two sourced observations gives a 13.91% increase in that USD equivalent, while the quoted USD/CHF rate falls 12.21%. Display values are rounded; the formula preserves the exact inputs.

This is an exchange-rate translation, not a portfolio simulation. No loan amount, margin requirement, financing cost or asset gain is assumed. Liquidation risk requires those inputs and the lender’s contractual rules.

> **WHAT THE CHART CANNOT SHOW**
>
> The H.10 snapshot does not measure the worst intraday executable rate, spread widening or a particular investor’s fill. Do not substitute it for a stress-test loss estimate.

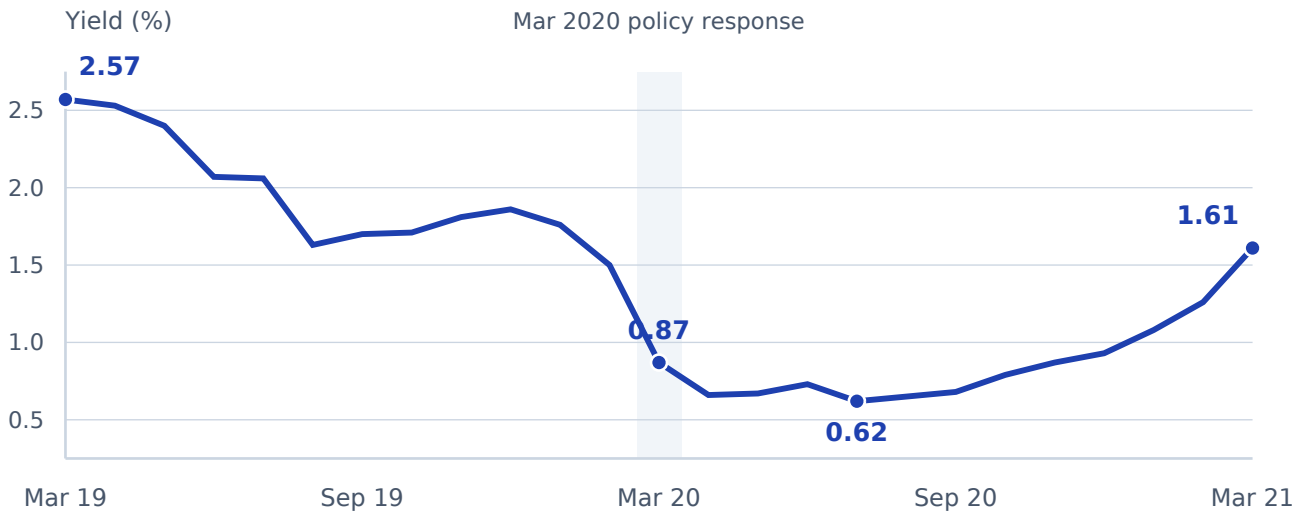
Sources: S23, S24. All original observations are retained in the companion CSV.

12 • WORKED CHART: COVID-19

A falling monthly yield does not prove daily liquidity

Broad yield direction and market functioning are different questions.

US 10-year Treasury yield | March 2019–March 2021



Source: FRED GS10 [S36]. Percent, monthly business-day averages. How to read: the line describes the broad yield environment around March 2020, not intramonth trading conditions. Non-zero scale; retrieved 5 November 2026.

Different policy milestones serve different annotation purposes

15 MARCH

Fed announces 0–0.25% target range

→

16 MARCH

New range becomes effective

→

23 MARCH

Purchases to support market functioning

Reported chronology [S26–S29]. Date boxes are not proportional to elapsed time.

Monthly averages are insufficient for a liquidity claim. To examine a dash-for-cash episode, add daily or intraday observations, bid–ask spreads and funding indicators where reliable data are available. This report deliberately does not invent an intramonth yield path from monthly values.

Keep the response separate from the trigger. The Sunday Fed statement is a policy milestone within a broader health and economic episode. It should not be coded as the date COVID-19 began, nor should later easing be treated as the sole explanation for each market movement.

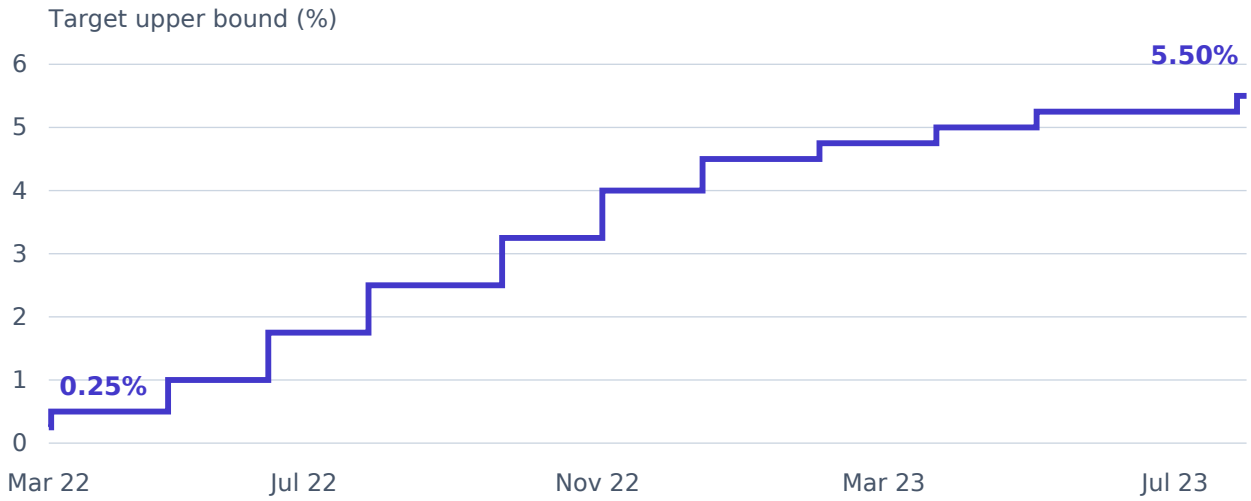
Sources: S26, S27, S29, S36.

13 • WORKED CHART: POLICY EFFECTIVE DATES

The 2022–23 tightening sequence raised the ceiling by 525 bp

The plotted series is the target range’s upper bound, not the effective federal funds rate.

Federal funds target upper bound | March 2022–July 2023



Source: Federal Reserve open-market-operations history [S29]. Unit: percent. How to read: steps change on effective dates; the baseline is 0.25%. The first decision was announced on 16 March and became effective on 17 March. Retrieved 5 November 2026.

Effective milestone	Target range	Upper bound change from baseline
17 March 2022	0.25–0.50%	+25 bp
16 June 2022	1.50–1.75%	+150 bp
15 December 2022	4.25–4.50%	+425 bp
4 May 2023	5.00–5.25%	+500 bp
27 July 2023	5.25–5.50%	+525 bp

The complete sequence matters. Ending the narrative in May 2023 misses the later rise to 5.25–5.50%. The difference in upper bounds is 5.50 − 0.25 = 5.25 percentage points, or 525 basis points. This historical endpoint says nothing about the current policy rate.

Do not mechanically translate policy tightening into a bond return. A specific bond’s outcome depends on its yield curve, coupon, maturity, spread, cash flows and holding period. A funding rate can also reset on a different schedule from the asset’s income.

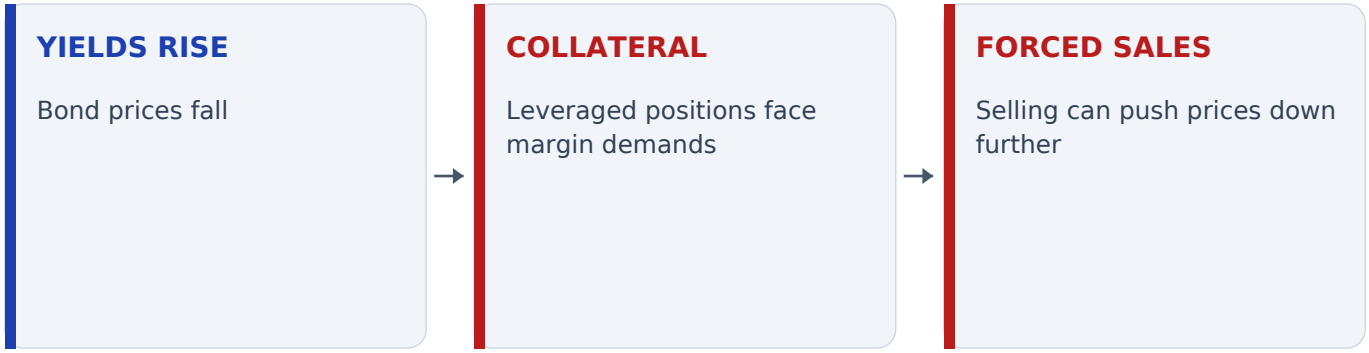
Sources: S28, S29. All effective-date steps and both target boundaries are in the companion CSV.

14 • WORKED CASE: UK GILTS AND LDI

The gilt crisis shows how leverage can amplify a price move

A safe issuer does not guarantee a safe financing structure.

A feedback loop, not just an initial shock



Conceptual diagram. LDI means liability-driven investment. Sources [S30]; no synthetic yield curve is shown.

Date	Dated milestone	Annotation type
23 September 2022	UK fiscal statement	Trigger / policy announcement
28 September 2022	Temporary conventional gilt purchases	Financial-stability intervention
11 October 2022	Index-linked gilts added	Programme expansion
14 October 2022	Temporary purchases end	Programme end

REPORTED MAGNITUDE

By 28 September, the Bank of England reports a 130-basis-point rise in 30-year gilt yields over three trading days. This is a reported period change, not a complete daily dataset.

The intervention's purpose matters. The BoE describes temporary, targeted purchases designed to restore functioning and allow LDI funds to build resilience. A chart label should distinguish that objective from a general monetary-policy easing programme.

Do not fabricate the missing line. This report links the BoE's original annotated chart, but has not extracted a full underlying gilt-yield dataset. A future numerical overlay should use verified observations and identify their licensing basis before redistribution.

Sources: S30. The chronology and reported magnitude are facts; the portfolio lesson is analysis. The source provides the original chart and operational details.

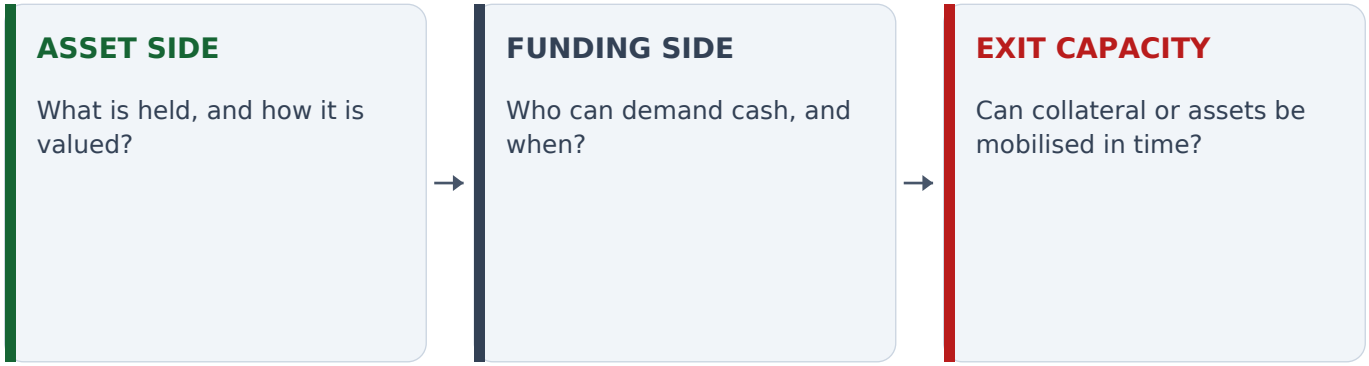
15 • COMPARATIVE CASE: 2023–2024

Bank runs and carry unwinds need separate diagnostic panels

A shared label such as “risk-off” is too vague to explain the mechanism.

Episode	Documented issue	What the chart should separate
SVB, 10 March 2023	Interest-rate risk, concentrated deposits, management and supervisory weaknesses [S31]	Asset valuation, deposit flight and available liquidity
Credit Suisse, 19 March 2023	Takeover and liquidity-support announcement [S32]	Announcement, next trading session and legal completion
August 2024 turbulence	Macro news and amplification through leveraged carry unwinding [S33]	Funding currency, equity exposure and deleveraging

The balance-sheet questions are more useful than the headline



Analytical comparison. No claim that all three episodes share one cause or severity.

Keep institutional evidence and market interpretation apart. The Fed’s SVB review documents weaknesses, while the SNB’s Credit Suisse release identifies a specific rescue announcement. These are not interchangeable evidence for a universal “bank failure” mechanism, and neither justifies assuming that every type of security receives the same protection.

Positioning data can be incomplete even when the market event is clear. BIS discusses carry unwinds as an amplifier in August 2024. The catalogue therefore includes the episode without claiming that total carry exposure, every participant’s leverage or the contribution to each price move is precisely observable.

IMPLEMENTATION ACTION

Separate source-backed facts from a proposed transmission chain. Show “analysis” on the chain and keep uncertain position-size estimates out of default chart labels.

Sources: S31, S32, S33.

16 • SINGAPORE CONTEXT

Give Singapore its own lens without inventing a new event

A global episode can have local effects while retaining the same canonical record.

Singapore lens	Relevant episode	Useful chart pairing
Domestic adjustment	1985 recession [S07]	GDP or activity with year-level context
Regional financing	Asian financial crisis [S10, S15]	Regional FX, funding and domestic activity
Services disruption	SARS [S14, S15]	Travel and activity indicators with the WHO alert
Global funding and demand	GFC [S15–S18]	Treasury yields, FX and local activity
Currency framework	MAS policy architecture [S42]	S$NEER concepts kept distinct from USD/SGD

A local view does not require a duplicate history



Design recommendation. Country tags identify context; they do not establish causation.

USD/SGD is not Singapore's monetary-policy target. MAS explains its framework through the trade-weighted S$NEER, rather than a single bilateral currency pair. A USD/SGD chart is useful for translating USD assets into SGD, but it cannot by itself measure a policy-band shift.

The investor's reporting currency must be explicit. For an SGD-based view of a USD asset financed in CHF, asset value, loan value, income and financing costs are separate quantities. Historical FX annotations can explain their exposure, but they cannot supply missing loan terms or calculate a complete portfolio return.

> **ACTIONABLE DEFAULT**
>
> Offer Singapore as a filter and a reporting-currency lens. Keep the original event ID, add local sources where available, and label missing S$NEER or local-market observations instead of substituting a different series.

Sources: S07, S10, S14, S15, S40, S42.

17 • CHART EXPERIENCE

Reveal detail as the chart narrows

The user should see the history without losing the data underneath it.

Progressive disclosure controls clutter



Proposed interaction behaviour, not an implemented website feature.

Control	Recommended behaviour	Failure to prevent
Tree selection	Select family, mechanism or episode; indicate partial selections	A parent checkbox silently selecting unrelated records
Facet filters	OR within a facet; AND across facets; show active chips	An event disappearing without an explanation
Marker collisions	Cluster by visible pixel space; expand on tap	Unreadable labels stacked on top of one another
Chart alignment	Use the market’s actual calendar; retain the original event date	Moving Sunday news to Monday without explaining it
Episode band	Show defined, sourced or explicitly curated bounds	Painting a precise recovery interval without a definition
Evidence drawer	Fact, analysis, units, timestamps, source and related events	A source link that supports only part of the popup

Touch interaction must do the same work as hover. On a phone, a tap should open a persistent card with a close control and visible source links. Use a labelled icon or short milestone code, not colour alone, and preserve keyboard focus in the eventual implementation.

Chart relevance should be explainable. A Treasury chart can prioritise rates, inflation and funding milestones. A CHF chart can prioritise currency-regime decisions. Allow an “all events” override, but show why the default filter included or excluded a record.

> **NO SILENT PRODUCT CHANGE**
>
> This report defines behaviour only. Integrating it into an existing site requires inspecting the current code, preserving data logic, testing mobile flows and updating the sitemap and tag map after authorised changes.

18 • EVENT-STUDY METHOD

Measure changes honestly before discussing causes

A before/during/after panel should answer a defined question, not choose the most dramatic endpoints.

Measure	Calculation or rule	Do not confuse it with
Yield change	(Ending yield % − starting yield %) × 100 = basis points	Bond price return
Price return	(Ending price / starting price − 1) × 100	Total return including distributions
Total return	Use a valid total-return series or include cash-flow reinvestment explicitly	A price-only line
FX translation	Write units first; invert the quote where necessary	Symmetric percentage changes
Study window	Default context: 12 months before / after where available; add daily zoom	A claim that the event caused the entire window
Benchmark comparison	Use aligned periods and define the comparator in advance	A controlled causal estimate

There is no single clean “event effect” in these descriptive charts. Expectations can change beforehand, other news arrives during the window, and policy responds to the same deterioration affecting prices. A descriptive overlay is valuable precisely when it makes these limitations visible rather than dressing them up as a model result.

Do not fill a long data gap just to complete the picture. DGS10 does not provide a 1929 daily history. DGS30 was discontinued in February 2002 and reintroduced in February 2006. Missing observations should remain gaps, with an alternative series introduced only under a separate label and methodology.

Point-in-time analysis needs point-in-time information. A historical backtest should use information available then, not revised macro data or an ex-post list of winning events. ALFRED provides vintages for suitable series; eligibility and release timing still need to be checked for each dataset.

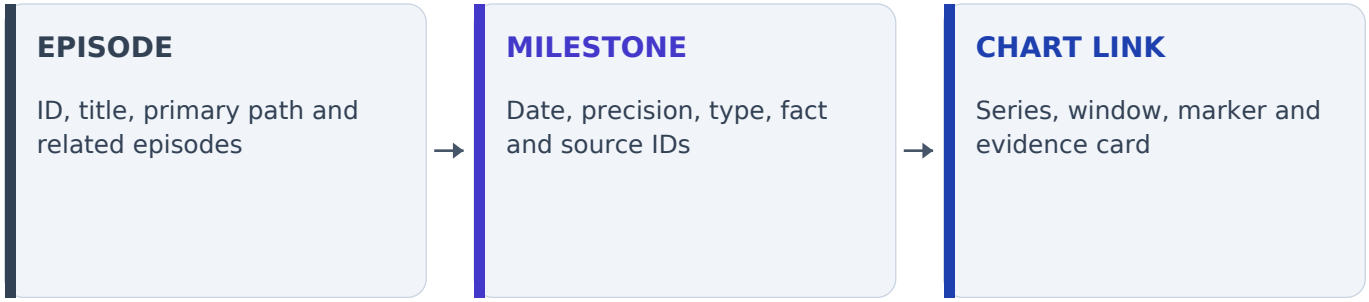
Sources: S36, S38, S41, S43. Method recommendations are analytical, not estimated causal results.

19 • DATA MODEL

Store the evidence separately from the display settings

Stable IDs preserve corrections, multiple charts and the user’s approved taxonomy.

The core model should survive a redesign



Proposed schema. One source can support several records; each claim still needs an appropriate reference.

```
episode_id: "GB-2022-LDI"
primary_path: ["liquidity", "Margin calls and forced deleveraging"]
milestone:
  id: "GB-2022-LDI-M02"
  date: "2022-09-28"
  date_precision: "day"
  type: "policy_intervention"
  fact: "Temporary gilt purchases announced"
  source_ids: ["S30"]
  announcement_time: null
  evidence_status: "REPORTED"
chart_link:
  observed_series: null
  show_source: true
```

Null is an honest state. The example does not fabricate an announcement time or an underlying daily gilt dataset. A populated marker can still be valid while a complete numerical overlay remains unavailable.

The companion JSON is a research starter, not a production migration. It contains 26 episode records, sourced milestones, a source register and six demonstration datasets. Canonical colour assignments should resolve through the supplied workbook. Proposed mechanism labels remain neutral until formally mapped.

GOVERNANCE FIELDS TO ADD AT INTEGRATION

Record schema version, last verification, editor, change reason, claim-level source locations, optional exact time zone, date-window basis and data licence. Keep an audit trail when an episode’s primary path changes.

Sources: S30.

20 • PRIORITY AND ACCEPTANCE TESTS

Build the reliable minimum before the impressive interface

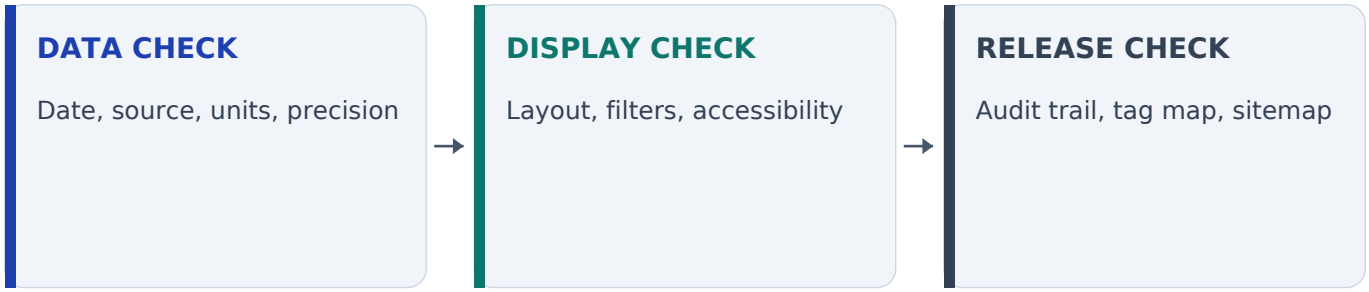
The first release should make errors easy to spot and difficult to repeat.

Priority	Deliverable	Acceptance test
1	Canonical records and source drawer	Every visible milestone has a supported date, type and source; unknown values remain explicit
2	Read-only chart overlay	Markers use the right calendar and units; a monthly mean is labelled as such
3	Tree and independent filters	Parent selection, partial state and AND/OR behaviour work; no duplicate IDs
4	Mobile and greyscale clarity	Labels survive narrow screens; touch and keyboard reveal all essential information
5	Event comparison and maintenance	Windows, benchmarks and revisions are visible; sitemap and tag map reflect authorised changes

The structural change is moderate if the existing charts already accept annotations. The reusable catalogue is separate from the chart renderer, which limits disruption. The work becomes larger if different pages maintain incompatible dates, data formats or filter state; that cannot be judged without inspecting the current codebase.

Validation should target realistic failure modes. Test a Sunday announcement, a month-only episode, a missing holiday observation, overlapping events, an inverted FX pair, an unavailable series and a source correction. These cases reveal more than a demonstration where every field is conveniently complete.

Quality gates are more useful than a vague confidence score



Proposed release sequence. No implementation or release is claimed in this report.

BOUNDARY

Do not automatically publish a corrected event or rerun a portfolio simulation merely because this research starter has changed. Research maintenance, website changes and investment calculations need distinct controls.

21 • STRATEGY, RISKS AND NEXT ACTIONS

The result should improve judgement, not manufacture certainty

The best historical overlay answers “what was happening?” before it claims “why did it move?”.

Use	Value	Main limitation
Investment discussion	Explain mechanisms behind a period of stress	Historical similarity does not establish the next outcome
Risk review	Separate rates, credit, FX, funding and liquidity	A chart cannot replace contractual margin and cash-flow inputs
Research navigation	Reach official evidence from the plotted date	Official sources can be retrospective and incomplete
Product design	Reuse one library across several chart types	The taxonomy still needs editorial ownership

> **RECOMMENDED FIRST RELEASE**
>
> Start with the six demonstrated contexts: the GFC, taper tantrum, CHF shock, COVID-19, Fed tightening and UK gilt stress. Retain the wider 26-episode catalogue as searchable context, with coverage and data availability shown explicitly.

Use history as a set of mechanisms, not as a set of promises. A funding mismatch, a policy-regime change and a valuation correction require different questions even when each is followed by a falling chart. The tree is valuable because it helps the reader distinguish them.

The most important risk is false precision. Exact-looking dates, return figures and recovery bands can be misleading when they mix periods, frequencies or definitions. This report preserves month and year precision, leaves unavailable inputs blank and identifies analytical choices rather than presenting them as historical facts.

The immediate action is data integration, not a visual redesign of everything. Inspect the existing chart interface, map its series IDs to the research starter, and validate a small read-only overlay. Add deeper comparisons only after source links, dates, units and mobile behaviour pass review.

Source → mechanism → decision

Ask what is documented, how it could transmit and which exposure should be checked. That sequence makes the historical library useful without turning it into a forecast.

Report status: complete selected-scope research and design deliverable. Coverage is not exhaustive. Website implementation, live quotes, portfolio-specific modelling and the two PowerPoint presentations are not part of this report file.

APPENDIX • SOURCE REGISTER 1/4

Official sources and reusable data links

Direct source links support the relevant claims; retrieval date is 5 November 2026.

S01 Stock Market Crash of 1929

Federal Reserve History. Crash milestones; distinguish a market crash from the subsequent banking crisis. Open official source ↗

S02 Banking Panics of 1931–33

Federal Reserve History. Banking crisis chronology and the 6 March 1933 national bank holiday. Open official source ↗

S03 Closing the Gold Window

Federal Reserve History. 15 August 1971 dollar–gold convertibility decision. Open official source ↗

S04 Oil Shock of 1973–74

Federal Reserve History. Oil embargo and supply-shock setting; not a single-day asset-return series. Open official source ↗

S05 Anti-Inflation Measures

Federal Reserve History. 6 October 1979 change in monetary-policy operating procedures. Open official source ↗

S06 Savings and Loan Crisis

Federal Reserve History. Interest-rate mismatch, deregulation, losses and resolution context. Open official source ↗

S07 The 1985 Recession

Singapore Government / SG101. Singapore recession context; use year precision rather than inventing a daily trigger. Open official source ↗

S08 Stock Market Crash of 1987

Federal Reserve History. 19 October crash and 20 October Fed liquidity response. Open official source ↗

S09 Operation of Monetary Policy, 1992 Q4

Bank of England. 16 September 1992 suspension of sterling ERM membership. Open official source ↗

S10 Asian Financial Crisis

Federal Reserve History. 2 July 1997 Thai baht event and regional credit/currency transmission. Open official source ↗

S11 Near Failure of Long-Term Capital Management

Federal Reserve History. 23 September 1998 private-sector recapitalisation; Fed facilitation was not Fed funding. Open official source ↗

APPENDIX • SOURCE REGISTER 2/4

Official sources and reusable data links

Direct source links support the relevant claims; retrieval date is 5 November 2026.

S12 Monetary Policy Report: Economic and Financial Developments

Federal Reserve Board. Contemporaneous discussion of the 2000 equity-market reversal. Open official source ↗

S13 September 11, 2001

Federal Reserve History. Operational and payment-system disruption, with emergency liquidity support. Open official source ↗

S14 WHO issues a global alert about cases of atypical pneumonia

World Health Organization. 12 March 2003 alert: a dated milestone, not the date the disease began. Open official source ↗

S15 Overcoming Multiple Crises

Singapore Government / SG101. Singapore context for the Asian crisis, SARS and the GFC. Open official source ↗

S16 Subprime Mortgage Crisis

Federal Reserve History. Housing, credit and funding mechanisms of the global financial crisis. Open official source ↗

S17 FDIC History: 2000–2009

Federal Deposit Insurance Corporation. September 2008 institutional milestones, including Lehman and AIG. Open official source ↗

S18 FOMC statement, 16 December 2008

Federal Reserve Board. Decision to establish a federal funds target range of 0–0.25%. Open official source ↗

S19 Technical features of Outright Monetary Transactions

European Central Bank. 6 September 2012 conditional OMT framework; do not equate announcement with purchases. Open official source ↗

S20 US Interest Rates and Emerging Market Currencies: Taking Stock 10 Years After the Taper Tantrum

Federal Reserve Board. May/June 2013 communication and international transmission. Open official source ↗

S21 SNB discontinues minimum exchange rate and lowers interest rate to −0.75%

Swiss National Bank. 15 January 2015 announcement; original source landing page. Open official source ↗

S22 Monetary policy assessment and SNB explanation

Swiss National Bank. Retrospective explanation of the EUR/CHF minimum exchange-rate decision. Open official source ↗

APPENDIX • SOURCE REGISTER 3/4

Official sources and reusable data links

Direct source links support the relevant claims; retrieval date is 5 November 2026.

S23 H.10 Foreign Exchange Rates, 20 January 2015 release

Federal Reserve Board. Daily observations for 12–16 January 2015: CHF and SGD per USD, USD per EUR. Open official source ↗

S24 H.10 Foreign Exchange Rates, 26 January 2015 release

Federal Reserve Board. Daily observations for 19–23 January 2015, including an ND holiday observation. Open official source ↗

S25 What Happened in Money Markets in September 2019?

Federal Reserve Board. Repo stress and the 17 September market intervention. Open official source ↗

S26 FOMC statement, 15 March 2020

Federal Reserve Board. Sunday policy announcement; separate the announcement from Monday effectiveness. Open official source ↗

S27 Federal Reserve announces extensive new measures to support the economy

Federal Reserve Board. 23 March purchase policy and measures to support market functioning. Open official source ↗

S28 FOMC statement, 16 March 2022

Federal Reserve Board. Announcement of the first increase of the 2022–23 tightening sequence. Open official source ↗

S29 Open Market Operations: federal funds target changes

Federal Reserve Board. Effective-date history of target-range changes. Dates must not be relabelled as announcement dates. Open official source ↗

S30 Financial stability buy/sell tools: a gilt market case study

Bank of England. 2022 fiscal trigger, gilt/LDI mechanism, purchase dates and the distinction from monetary easing. Open official source ↗

S31 Review of Federal Reserve Supervision and Regulation of SVB: Executive Summary

Federal Reserve Board. 10 March 2023 failure; management, liquidity, interest-rate and supervisory weaknesses. Open official source ↗

S32 SNB provides substantial liquidity assistance to support UBS takeover of Credit Suisse

Swiss National Bank. 19 March 2023 rescue announcement, not the later legal completion of the acquisition. Open official source ↗

S33 BIS Bulletin 90: The market turbulence and carry trade unwind of August 2024

Bank for International Settlements. Primary research, especially pages 1–3; macro news, leverage and the 5 August episode. Open official source ↗

APPENDIX • SOURCE REGISTER 4/4

Official sources and reusable data links

Direct source links support the relevant claims; retrieval date is 5 November 2026.

S34 Global Financial Stability Report, April 2025

International Monetary Fund. Tariff-related financial conditions and market volatility; this is an April report, not a full subsequent chronology. Open official source ↗

S35 The Global Economy Enters a New Era

International Monetary Fund. 2 April 2025 tariff announcement and distinction between policy uncertainty and observed outcomes. Open official source ↗

S36 GS10: 10-Year Treasury Constant Maturity Rate, monthly

FRED / Federal Reserve Board. Monthly business-day averages, percent; underlying observations for three report charts. Open official source ↗

S37 EXSZUS: Switzerland / US Foreign Exchange Rate, monthly

FRED / Federal Reserve Board. Monthly average CHF per USD. Original four-decimal observations retained. Open official source ↗

S38 DGS10: 10-Year Treasury Constant Maturity Rate, daily

FRED / Federal Reserve Board. Daily series for future implementation; history begins in 1962, not 1929. Open official source ↗

S39 DEXSZUS: Switzerland / US Foreign Exchange Rate, daily

FRED / Federal Reserve Board. Daily CHF per USD; use quote units explicitly. Open official source ↗

S40 DEXSIUS: Singapore / US Foreign Exchange Rate, daily

FRED / Federal Reserve Board. SGD per USD. A bilateral exchange rate is not Singapore's policy basket. Open official source ↗

S41 DGS30: 30-Year Treasury Constant Maturity Rate, daily

FRED / Federal Reserve Board. Series discontinued 18 February 2002; reintroduced 9 February 2006. Preserve the gap. Open official source ↗

S42 Singapore's Monetary Policy Framework

Monetary Authority of Singapore. Trade-weighted S$NEER exchange-rate framework, distinct from bilateral FX. Open official source ↗

S43 ALFRED: Archival Federal Reserve Economic Data

Federal Reserve Bank of St Louis. Point-in-time data vintages for later event studies and backtests. Open official source ↗

S44 Financial Crisis Inquiry Report

US Government Publishing Office / FCIC. Official long-form further reading, including dissenting views. Not reproduced here. Open official source ↗